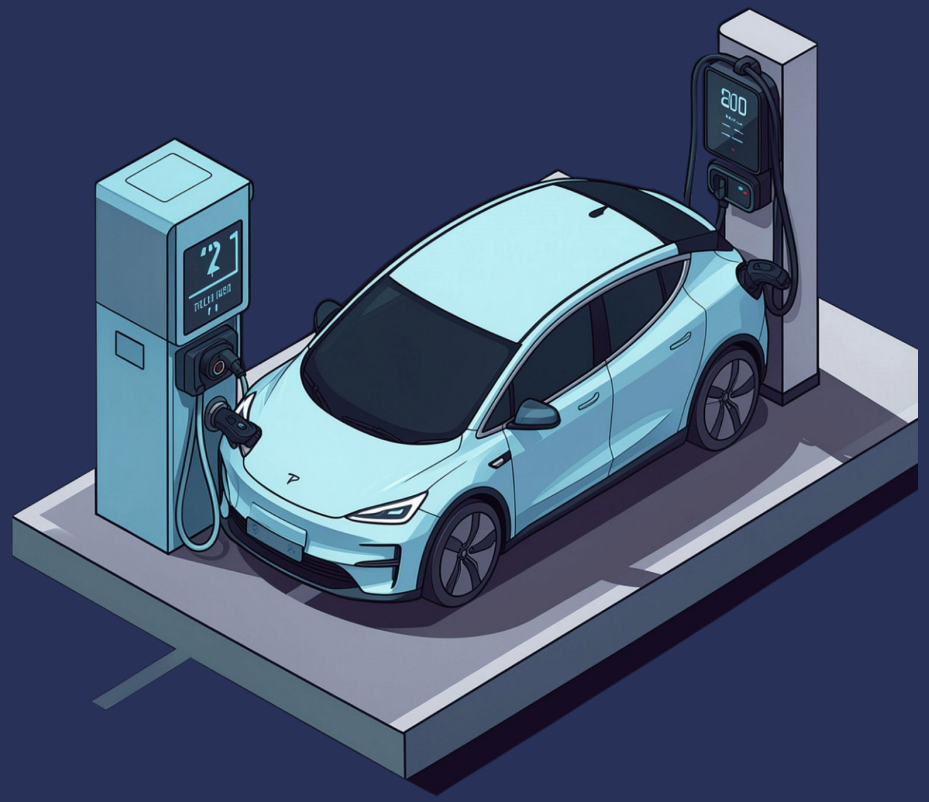


Cloud-Native EV Charging Station Reservation with Wait-Time Guardian

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Problem Statement

- Growing EV adoption is increasing pressure on public charging infrastructure.
- EV users frequently face **charging-station congestion and unpredictable waiting times**.
- Existing systems mainly provide static station information.
- Users may not know whether a charging slot will actually be available when they arrive.
- Long waiting times can cause:
  - Wasted travel time
  - Changes in travel plans
  - Increased range anxiety
  - Inefficient charging-station utilization

Need:
An intelligent and scalable system that combines **real-time reservation management with dynamic wait-time estimation**.

System Requirements

Functional Requirements

- Charging-station information management
- Charging-slot reservation
- Reservation-load management
- Real-time wait-time estimation
- Analysis of current and upcoming reservations
- Identification of comparatively less-congested stations
- Containerized service deployment
- Kubernetes-based orchestration
- Automatic scaling based on demand

Non-Functional Requirements

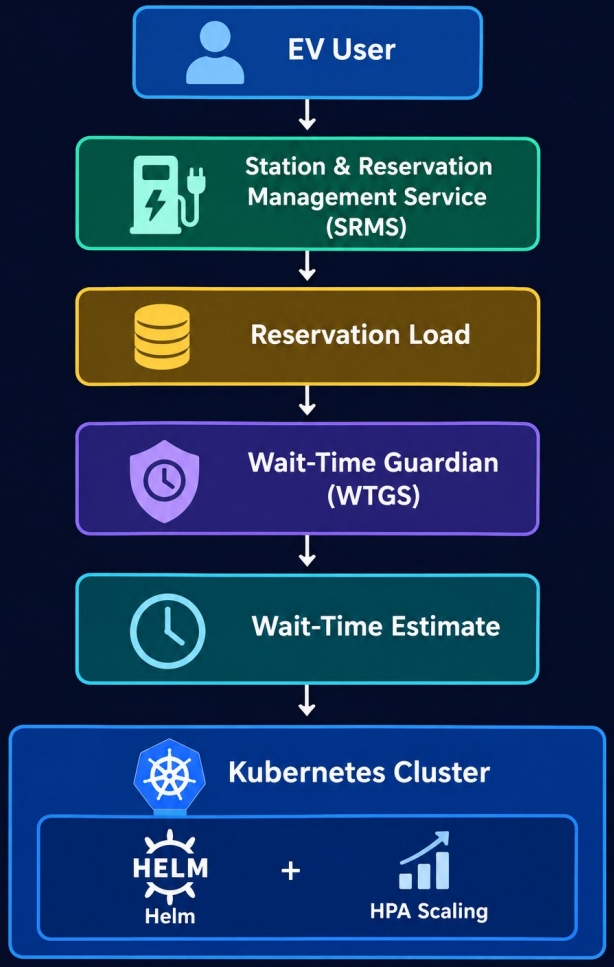
- Scalability
- Reliability
- Availability
- Responsiveness
- Fault tolerance
- Maintainability

System Architecture

Architecture Type: Cloud-Native Microservice Architecture.

Main Components:
1. SRMS
Station and Reservation Management Service
2. WTGS
Wait-Time Guardian Service
3. HPADF
Helm-Packaged Autoscaling Deployment Framework

Infrastructure:
Docker Containers
Kubernetes Cluster
Helm
Horizontal Pod Autoscaler (HPA)



Technology Stack

Application Layer

- Microservices
- REST-based service communication

Containerization

- Docker
- Orchestration
- Kubernetes

Deployment & Packaging

- Helm
- Autoscaling
- Kubernetes Horizontal Pod Autoscaler (HPA)

Testing

- Simulated reservation load
- Concurrent reservation requests
- Scaling behavior testing

Station and Reservation Management Service

Manages charging stations and charging-slot reservations.

KEY FUNCTIONS

- Manage station information
- Manage charging slots
- Process reservation requests
- Handle concurrent reservation requests
- Maintain reservation information
- Provide reservation-load information

PURPOSE

Manages charging stations and charging-slot reservations.



Station & Reservation Management Service

IMPLEMENTATION

- Independent microservice
- Containerized using Docker
- Deployed using Kubernetes
- Tested with concurrent reservation requests

OUTCOME

- Reliable reservation processing without service disruption.

Wait-Time Guardian Service

Provides dynamic waiting-time estimates for EV charging stations.

Purpose

Provides dynamic waiting-time estimates for EV charging stations.

Key Functions

- Analyze current reservation load
- Analyze upcoming reservations
- Estimate waiting time
- Provide dynamic wait-time information
- Identify comparatively less-congested stations

Working

Current Reservations + Upcoming Reservations

↓

Reservation Load Analysis

↓

Wait-Time Estimation

↓

User Guidance

Outcome



Responsive wait-time estimation aligned with simulated reservation load.

Helm-Packaged Autoscaling Deployment Framework

Provides reproducible deployment and automatic scaling.

Purpose

Provides reproducible deployment and automatic scaling.

Key Technologies

- Docker
- Kubernetes
- Helm
- Horizontal Pod Autoscaler

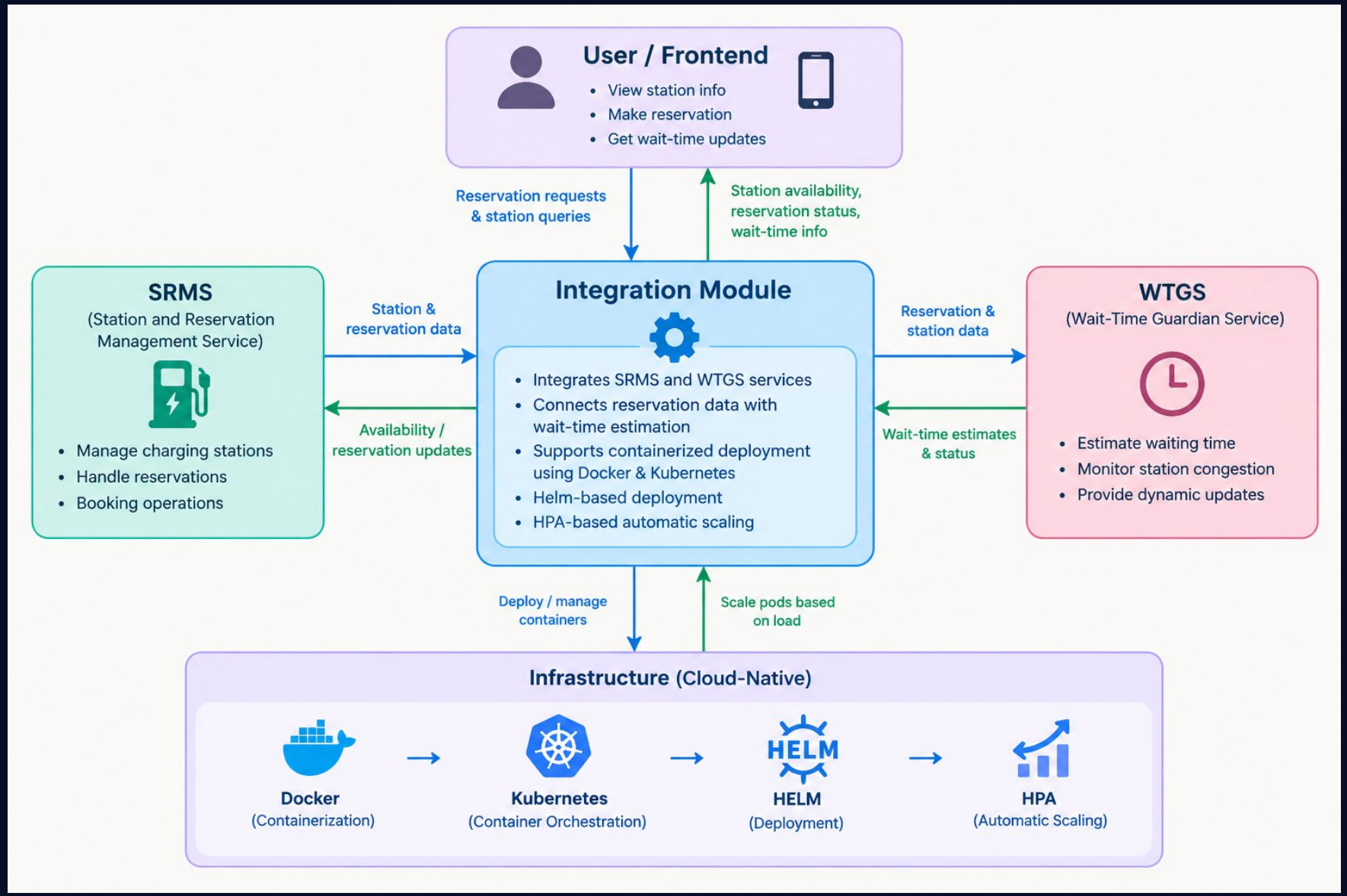
Benefits

- Reproducible deployment
- Demand-based scaling
- Improved resilience
- Simplified operational workflow

Working



Integration Module



Integration Module

- Integrates SRMS and WTGS services
- Connects reservation data with wait-time estimation
- Supports containerized deployment using Docker & Kubernetes
- Helm-based deployment
- HPA-based automatic scaling

Thank You